

UniBrain: An Ontology-Driven Academic Reasoning System

UniBrain is not a traditional language-model-powered chatbot. Its core value lies in the deterministic connection between structured university data, student-specific context, and rule-based academic reasoning. The language model serves as an explanatory interface, not a decision-maker.

1. University Context as a Modeled Ontology

University knowledge in UniBrain is represented through explicit entities and relationships, including campuses, majors, requirements, requirement groups, slots, bundles, and course options. This information is modeled as structured data rather than unstructured text. As a result, UniBrain operates over verifiable institutional truth instead of probabilistic interpretation.

2. Student Context as Runtime State

Each student interacts with UniBrain through a StudentProfile that captures their academic state, such as declared major, campus, completed courses, in-progress coursework, and expected graduation timeline. This context directly influences all reasoning outcomes, allowing UniBrain to answer questions specific to the individual student rather than providing generic responses.

3. Deterministic Academic Reasoning

UniBrain evaluates academic progress using deterministic logic implemented through database queries and application-level rules. Requirement satisfaction, missing coursework, and conflicts are computed explicitly by the system. The language model is restricted to translating these results into human-readable explanations.

System Architecture Philosophy

UniBrain functions as a rules-driven expert system augmented with a natural language interface. The language model does not invent facts or academic decisions. Instead, it explains outcomes derived from the ontology and student context in a clear, conversational manner.

Why This Matters

This architecture ensures auditability, consistency, and trust. Identical inputs always yield identical results, enabling advisors and administrators to verify UniBrain's reasoning. Improvements can be made by evolving the ontology or rules without retraining or fine-tuning language models.